MTH316 Specimen

Jan 2021

Donaldson Tan

MTH316

Specimen Examination Paper

Multivariable Calculus

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Specimen Examination Paper contains **SIX (6)** questions and comprises **FOUR (4)** pages (including cover page).

2. You must answer **SIX (6)** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

5. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

6. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

7. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Question 1

A function of two variables (x, y) is given as:

$$f(x,y) = \begin{cases} \dfrac{3x^{\mathrm{m}}y}{3x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ \text{n if } (x,y) = (0,0) \end{cases}$$

m, n are integer.

(a) If $m = 1$, determine the existence of $\lim\limits_{(x,y)\to(0,0)} f(x,y)$.

(5 marks)

(b) If $m = 2$, determine the existence of $\lim\limits_{(x,y)\to(0,0)} f(x,y)$.

(6 marks)

(c) If $m = 2$, show that $f(x,y)$ is continuous or not at $(0,0)$.

(4 marks)

Question 2

Suppose you are climbing a hill whose shape is given by the function of

$$z = 200 - 0.01x^2 - 0.02y^2,$$

where x, y, and z are measured in meters, and you are standing at a point with coordinates (20, -30, 178). The positive x-axis points east and the positive y-axis points north.

(a) Find the gradient of z at (20, -30, 178).

(5 marks)

(b) If you walk toward North, will you start to ascend or descend? At what rate?

(5 marks)

(c) If you walk southwest, will you start to ascend or descend? At what rate?

(5 marks)

(d) If you start to walk, in which direction is the slope largest? What is the rate of ascent in that direction?

(5 marks)

Question 3

(a) The total production z of a certain product depends on the amount x of labour used and the amount y of capital investment which is given by the model $z = bx^a y^{1-a}$ follows from certain economic assumptions, where b and a are positive constants and a< 1. If the cost of a unit of labour is m and the cost of a unit of capital is n, and the company can spend only p dollars as its total budget, then maximizing the production p is subject to the constraint $mx + ny = p$.
Apply Lagrange multipliers to find the value x and y that the maximum production z.

(10 marks)

(b) We now suppose that the production is fixed at $bx^a y^{1-a} = q$, where q is a constant. What values of x and y minimize the cost function $f(x, y) = mx + ny$.

(10 marks)

Question 4

Find the volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$ and above the cone $z = \sqrt{x^2 + y^2}$.

(10 marks)

Question 5

If $\mathbf{F} = y^2 cosx\ \mathbf{i} + (x^2 + 2ysinx)\mathbf{j}$, where C is the triangular curve consisting of the line segments from $(0,0)$ to $(2,6)$, from $(2,6)$ to $(2,0)$, and from $(2,0)$ to $(0,0)$.

(a) Define curve C_1 is a line from $(0,0)$ to $(2,6)$, C_2 is a line from $(2,6)$ to $(2,0)$, and C_3 is a line from $(2,0)$ to $(0,0)$. Compute the line integral of $\mathbf{F}$ along $C_1, C_2, C_3, \int_{C_1} \mathbf{F} \cdot d\mathbf{r}, \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ and $\int_{C_3} \mathbf{F} \cdot d\mathbf{r}$.

(10 marks)

(b) Find the curl of $\mathbf{F}$.

(5 marks)

(c) Suppose D is enclosed by C_1, C_2 and C_3. Use Green's Theorem to evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

(5 marks)

Question 6

Let $\mathbf{F}$ be the vector field $\mathbf{F}(x, y, z) = -y^2\mathbf{i} + x\mathbf{j} + z^2\mathbf{k}$ and $\mathbf{S}$ be the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 1$, oriented upward and C is the intersection of the cylinder $x^2 + y^2 = 1$ with the plane $z = 1$, oriented in the counter clockwise direction as viewed from above.

(a) Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

(7 marks)

(b) Evaluate curl $\mathbf{F}$.

(4 marks)

(c) Use Stokes' Theorem to compute the integral $\iint_S \text{curl}\, \mathbf{F} \cdot d\mathbf{S}$.

(4 marks)

----- END OF PAPER -----

1.

$$f(x,y) = \begin{cases} \dfrac{3x^m y}{3x^2+y^2} & \text{if } (x,y) \neq 0 \\ n & \text{if } (x,y) = (0,0) \end{cases}$$

(a) consider path along x-axis

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(x,0)$$

$$= \lim_{x\to 0} \frac{3(0)y}{3(0)^2+y^2}$$

$$= 0$$

consider path along $y=x$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(x,x)$$

$$= \lim_{x\to 0} \frac{3x(x)}{3x^2+x^2}$$

$$= \lim_{x\to 0} \frac{3x^2}{4x^2}$$

$$= \frac{3}{4}$$

$\therefore \lim_{(x,y)\to(0,0)} f(x,y)$ doesn't exist.

(b) consider path along x-axis

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(x,0)$$

$$= \lim_{x\to 0} \frac{3x^2(0)}{3x^2+0^2}$$

$$= 0$$

consider path along $y=x$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(x,x)$$

$$= \lim_{x\to 0} \frac{3x^2(x)}{3x^2+x^2}$$

$$= \lim_{x\to 0} \frac{3x^3}{4x^2}$$

$$= \lim_{x\to 0} \frac{3}{4}x$$

$$= 0$$

suppose $\lim_{(x,y)\to(0,0)} f(x,y) = 0$,

for any $\varepsilon > 0$,

$|f(x,y)-0| < \varepsilon \Rightarrow \exists \delta > 0$ such that $\sqrt{(x-0)^2+(y-0)^2} < \delta$

$$|f(x,y)-0| < \varepsilon$$

$$\left|\frac{3x^2y}{3x^2+y^2}\right| < \varepsilon$$

$$\frac{3x^2|y|}{3x^2+y^2} < \varepsilon$$

$|x| \leq \sqrt{x^2+y^2}$, $|y| \leq \sqrt{x^2+y^2}$

$3x^2|y| = 3|x|^2|y| \leq 3(x^2+y^2)\sqrt{x^2+y^2}$

$$\frac{3x^2|y|}{3x^2+y^2} = \frac{3|x|^2|y|}{3|x|^2+|y|^2} \leq \frac{3(x^2+y^2)\sqrt{x^2+y^2}}{3(x^2+y^2)+(x^2+y^2)} = \frac{3(x^2+y^2)\sqrt{x^2+y^2}}{4(x^2+y^2)} < \varepsilon$$

$$\frac{3}{4}\sqrt{x^2+y^2} < \varepsilon$$

$$\sqrt{x^2+y^2} < \frac{4}{3}\varepsilon$$

By Density Property,

$\exists \delta$ s.t. $\sqrt{x^2+y^2} < \delta < \frac{4}{3}\varepsilon$

Hence, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$

1(c) For $f(x,y)$ to be continuous at $(0,0)$,

$$\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0)$$

However, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$

$$f(0,0) = n$$

If $n=0$, then $f(x,y)$ is continuous at $(0,0)$

2(a) $z = 200 - 0.01x^2 - 0.02y^2$

$\frac{\partial z}{\partial x} = -0.02x$ $\quad \nabla z = \begin{bmatrix} -0.02x \\ -0.04y \end{bmatrix}$

$\frac{\partial z}{\partial y} = -0.04y$

$$\nabla z(20,-30,178) = \begin{bmatrix} -0.02(20) \\ -0.04(-30) \end{bmatrix} = \begin{bmatrix} -0.4 \\ 1.2 \end{bmatrix}$$

(b) $u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$D_u(x,y) = \nabla z \cdot u = \begin{bmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{\partial z}{\partial y} = -0.04y$$

$$D_u(20,-30) = -0.04(-30) = 1.2$$

$\therefore$ rate of ascent $= 1.2$

2(c) $|u| = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

$$\frac{u}{|u|} = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$D_u(x,y) = \nabla z \cdot \frac{u}{|u|} = \begin{bmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} -0.02x \\ -0.04y \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}}\left[0.02x + 0.04y\right]$$

$$D_u(20,-30) = \frac{1}{\sqrt{2}}\left[0.02(20) + 0.04(-30)\right] = \frac{1}{\sqrt{2}}(0.4 - 1.2) = \frac{1}{\sqrt{2}}(-0.8) = -\frac{4}{5\sqrt{2}} = -\frac{2\sqrt{2}}{5}$$

$\therefore$ rate of descent $= \frac{2\sqrt{2}}{5}$

2(d)

$$\nabla z(20,-30) = \begin{bmatrix} -0.02(20) \\ -0.04(-30) \end{bmatrix} = \frac{2}{5}\begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$\therefore$ slope largest in $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ direction

$$\left|\begin{bmatrix} -1 \\ 3 \end{bmatrix}\right| = \sqrt{10}$$

$$\Rightarrow u = \frac{1}{\sqrt{10}}\begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$D_u(x,y) = \begin{bmatrix} -0.02x \\ -0.04y \end{bmatrix} \cdot \frac{1}{\sqrt{10}}\begin{bmatrix} -1 \\ 3 \end{bmatrix} = \frac{1}{\sqrt{10}}(0.02x - 0.12y)$$

$$D_u(20,-30) = \frac{1}{\sqrt{10}}(0.4 + 3.6) = \frac{4}{\sqrt{10}} = \frac{4}{\sqrt{10}} = \frac{2}{\sqrt{5}}$$

rate of ascent $= \frac{2}{\sqrt{5}}$

3(a) maximize $z(x,y)$

subject to $g(x,y) = p$

where : $z(x,y) = b x^a y^{1-a}$,

$g(x,y) = mx + ny$

$b > 0$, $0 < a < 1$

$$\nabla z = \lambda \nabla g$$

$$\Rightarrow \frac{\partial z}{\partial x} = \lambda \frac{\partial g}{\partial x}$$

$$\Rightarrow \frac{\partial z}{\partial y} = \lambda \frac{\partial g}{\partial y}$$

$$\frac{\partial z}{\partial x} = abx^{a-1}y^{1-a}$$

$$\frac{\partial z}{\partial y} = (1-a)bx^a y^{-a}$$

$$\frac{\partial g}{\partial x} = m$$

$$\frac{\partial g}{\partial y} = n$$

$$abx^{a-1}y^{1-a} = \lambda m \quad -(1)$$

$$(1-a)bx^a y^{-a} = \lambda n \quad -(2)$$

$$mx + ny = p \quad -(3)$$

$$\lambda = \frac{abx^{a-1}y^{1-a}}{m} = \frac{(1-a)bx^a y^{-a}}{n}$$

$$\frac{ab}{m}\left(\frac{x}{y}\right)^{a-1} = \frac{(1-a)b}{n}\left(\frac{x}{y}\right)^a$$

$$\frac{ab}{m}\left(\frac{x}{y}\right)^{a-1} - \frac{(1-a)b}{n}\left(\frac{x}{y}\right)^a = 0$$

$$b\left(\frac{x}{y}\right)^a\left[\frac{a}{m}\left(\frac{y}{x}\right) - \frac{1-a}{n}\right] = 0$$

case $\left(\frac{x}{y}\right)^a = 0$:

$x = 0$

(reject, because $z(0,y) = 0$)

case $\frac{a}{m}\left(\frac{y}{x}\right) - \frac{1-a}{n} = 0$

$$\frac{a}{m}\left(\frac{y}{x}\right) = \frac{1-a}{n}$$

$$\frac{y}{x} = \frac{m(1-a)}{na}$$

$$\frac{ny}{x} = \frac{m(1-a)}{a}$$

$$\frac{p - mx}{x} = \frac{m(1-a)}{a}$$

$$\frac{p}{x} - m = \frac{m}{a} - m$$

$$\frac{p}{x} = \frac{m}{a}$$

$$\frac{x}{p} = \frac{a}{m}$$

$$\therefore x = \frac{ap}{m}$$

$$\therefore y = \frac{p - mx}{n}$$

$$= \frac{p - ap}{n}$$

$$= \frac{(1-a)p}{n}$$

3(b) minimise $f(x,y)$

subject to $z(x,y) = q$

where:

$$f(x,y) = mx + ny$$
$$z(x,y) = bx^a y^{1-a}$$

$$\nabla f = \lambda \nabla z$$

$$\Rightarrow \frac{\partial f}{\partial x} = \lambda \frac{\partial z}{\partial x}$$

$$\Rightarrow \frac{\partial f}{\partial y} = \lambda \frac{\partial z}{\partial y}$$

$$\frac{\partial f}{\partial x} = m, \quad \frac{\partial f}{\partial y} = n$$

$$\frac{\partial z}{\partial x} = abx^{a-1}y^{1-a}$$

$$\frac{\partial z}{\partial y} = (1-a)bx^a y^{-a}$$

$$m = \lambda abx^{a-1}y^{1-a}$$
$$n = \lambda (1-a)bx^a y^{-a}$$
$$q = bx^a y^{1-a}$$

$$\frac{1}{\lambda} = \frac{abx^{a-1}y^{1-a}}{m} = \frac{(1-a)bx^a y^{-a}}{n}$$

$$\frac{abx^{a-1}y^{1-a}}{m} - \frac{(1-a)bx^a y^{-a}}{n} = 0$$

$$b\left(\frac{x}{y}\right)^a \left[\frac{a}{m}\left(\frac{y}{x}\right) - \frac{(1-a)}{n}\right] = 0$$

case $\left(\frac{x}{y}\right)^a = 0$:

$\therefore x = 0$

(reject because $z(x,y)=q$)

case $\frac{a}{m}\left(\frac{y}{x}\right) - \frac{(1-a)}{n} = 0$

$$\frac{a}{m}\left(\frac{y}{x}\right) = \frac{(1-a)}{n}$$

$$\frac{y}{x} = \frac{m}{a} \cdot \frac{1-a}{n}$$

$$\frac{x}{y} = \frac{na}{m(1-a)}$$

$$q = bx^a y^{1-a}$$
$$= by\left(\frac{x}{y}\right)^a$$
$$= by\left[\frac{na}{m(1-a)}\right]^a$$

$$\therefore y = q \div b\left[\frac{na}{m(1-a)}\right]^a$$
$$= \frac{q}{b}\left[\frac{m(1-a)}{na}\right]^a$$

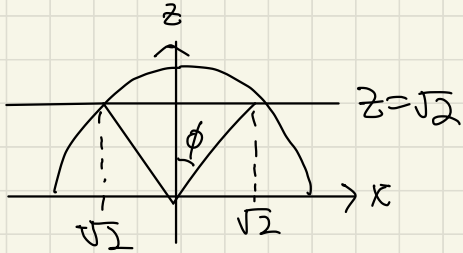
$$\therefore x = \left[\frac{m(1-a)}{na}\right]^{-1} y$$
$$= \frac{q}{b}\left[\frac{m(1-a)}{na}\right]^{a-1}$$

4. consider spherical coordinates

$x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$



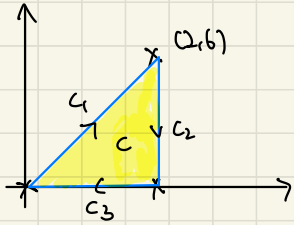
Find intersection between cone & sphere

$$x^2+y^2+z^2 = 2^2$$
$$x^2+y^2 = z^2$$
$$2z^2 = 2^2$$
$$z^2 = 2$$
$$\therefore z = \sqrt{2}$$

$$\tan\phi = \frac{\sqrt{2}}{\sqrt{2}} = 1$$
$$\therefore \phi = \tan^{-1}1 = \frac{\pi}{4}$$

$$\therefore \text{volume} = \int_0^{2\pi}\int_0^{\frac{\pi}{4}}\int_0^2 \rho^2\sin\phi \, d\rho \, d\phi \, d\theta$$
$$= \int_0^{2\pi}\int_0^{\frac{\pi}{4}} \left[\tfrac{1}{3}\rho^3\right]_0^2 \sin\phi \, d\phi \, d\theta$$
$$= \int_0^{2\pi}\int_0^{\frac{\pi}{4}} \tfrac{8}{3}\sin\phi \, d\phi \, d\theta$$
$$= \int_0^{2\pi} \tfrac{8}{3}\left[-\cos\phi\right]_0^{\frac{\pi}{4}} d\theta$$
$$= \cancel{\int_0^{2\pi} \tfrac{8}{3}(1-\sqrt{2})\, d\theta} \quad \int_0^{2\pi} \tfrac{8}{3}\left(1-\tfrac{\sqrt{2}}{2}\right) d\theta$$
$$= \cancel{\tfrac{8}{3}(1-\sqrt{2})[\theta]_0^{2\pi}} \quad \tfrac{8}{3}\left(1-\tfrac{\sqrt{2}}{2}\right)[\theta]_0^{2\pi}$$
$$= \cancel{\tfrac{16\pi}{3}(1-\sqrt{2})} \quad \tfrac{16\pi}{3}\left(1-\tfrac{\sqrt{2}}{2}\right)$$

5. $F(x,y) = \begin{bmatrix} y^2 \cos x \\ x^2 + 2y \sin x \end{bmatrix}$



take note C is clockwise, not anti clockwise

(a) C_1: $r(x) = \begin{bmatrix} x \\ 3x \end{bmatrix}$, $x = 0..2$, $r'(x) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$$\int_{C_1} F \cdot dr = \int_0^2 F \cdot r'(x) \cdot dx$$

$$= \int_0^2 \begin{bmatrix} y^2 \cos x \\ x^2 + 2y \sin x \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} dx$$

$$= \int_0^2 \begin{bmatrix} 9x^2 \cos x \\ x^2 + 6x \sin x \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} dx$$

$$= \int_0^2 9x^2 \cos x + 3x^2 + 18x \sin x \, dx$$

$$= \int_0^2 \underbrace{9x^2}_{u} \underbrace{\cos x \, dx}_{dv} + \int_0^2 18x \sin x \, dx + \int_0^2 3x^2 \, dx$$

$$= \left[9x^2 \sin x\right]_0^2 - \int_0^2 18x \sin x \, dx + \int_0^2 18x \sin x \, dx + \left[x^3\right]_0^2$$

$$= 36 \sin 2 + 8$$

C_2: $r(y) = \begin{bmatrix} 2 \\ y \end{bmatrix}$, $y = 6..0$, $r'(y) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\int_{C_2} F \cdot dr = \int_6^0 F(2,y) \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} dy$$

$$= \int_6^0 \begin{bmatrix} y^2 \cos 2 \\ 4 + 2y \sin 2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} dy$$

$$= \int_6^0 4 + 2y \sin 2 \, dy$$

$$= \left[4y + y^2 \sin 2\right]_6^0$$

$$= -24 - 36 \sin 2$$

5(a) C_3: $r(x) = \begin{bmatrix} x \\ 0 \end{bmatrix}$, $x = 2..0$, $r'(x) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\int_{C_3} F \cdot dr = \int_2^0 F(x,0) \cdot r'(x)\, dx$$
$$= \int_2^0 \begin{bmatrix} 0 \\ x^2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} dx$$
$$= \int_2^0 0\, dx$$
$$= 0$$

(b) $\text{curl } F = |\nabla \times F|$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x & x^2 + 2y \sin x & 0 \end{vmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ (2x + 2y\cos x) - 2y\cos x \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 2x \end{bmatrix}$$

(c) $P(x,y) = y^2 \cos x$

$Q(x,y) = x^2 + 2y \sin x$

Greens Theorem

$$\int_C F \cdot dr = -\iint_D \boxed{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}\, dA$$

because C is clockwise, not anti-clockwise

$$= -\iint_D 2x + 2y\cos x - 2y\cos x\, dA$$
$$= -\iint_D 2x\, dA$$
$$= -\int_0^2 \int_0^{3x} 2x\, dy\, dx$$
$$= -\int_0^2 2x[y]_0^{3x}\, dx$$
$$= -\int_0^2 6x^2\, dx = -\left[2x^3\right]_0^2 = -16$$

6. $F(x,y,z) = \begin{bmatrix} -y^2 \\ x \\ z^2 \end{bmatrix}$

(a) $C: r(\theta) = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 1 \end{bmatrix}, \ 0 \le \theta \le 2\pi$

$r'(\theta) = \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix} \quad F(\theta) = \begin{bmatrix} -\sin^2\theta \\ \cos\theta \\ 1 \end{bmatrix}$

$$\int_C F \cdot dr = \int_0^{2\pi} F(\theta) \cdot r'(\theta) \cdot d\theta$$

$$= \int_0^{2\pi} \begin{bmatrix} -\sin^2\theta \\ \cos\theta \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix} \cdot d\theta$$

$$= \int_0^{2\pi} \sin^3\theta + \cos^2\theta \, d\theta$$

$$= \int_0^{2\pi} \sin\theta(1-\cos^2\theta) + \cos^2\theta \, d\theta$$

$$= \int_0^{2\pi} \sin\theta - \sin\theta\cos^2\theta + \tfrac{1}{2}(\cos 2\theta + 1) \, d\theta$$

$$= \int_0^{2\pi} \sin\theta - \sin\theta\cos^2\theta + \tfrac{1}{2}\cos 2\theta + \tfrac{1}{2} \, d\theta$$

$$= \left[-\cos\theta + \tfrac{1}{3}\cos^3\theta + \tfrac{1}{4}\sin 2\theta + \tfrac{1}{2}\theta \right]_0^{2\pi}$$

$$= \pi$$

(b) $\text{curl} F = |\nabla \times F|$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & x & z^2 \end{vmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1+2y \end{bmatrix}$$

(c). By Stokes Theorem,

$$\iint_S \text{curl} F \cdot dS = \int_C F \cdot dr$$

$$= \pi$$